



Alibaba's robot delivery network at work (Credit: <http://emag.directindustry.com>)

to decrease costs besides improving reliability and stability.

Connecting consumers to connected systems

Connected systems can detect flaws and communicate/automatically make amendments to prevent faults in the final product. By connecting consumers to the entire lifecycle of a product, companies can observe a significantly higher adoption rate. It is important to create transparency in the value-chain as consumers do not differentiate between manufacturers and resellers. Since the same high-tech product is available from several companies, consumers nowadays value their experience with the brand over product design.

One of the best examples is, probably, Tesla. According to a study conducted by Experian, Tesla owners showed a higher loyalty rate to the company than electric vehicles (EV) in general. Also, many people who had pre-ordered Tesla Model 3 in 2016 were later on informed that delivery would be delayed due to some manufacturing issues. But because of the company's transparency regarding challenges, customers were willing to wait for years to get their vehicle.

Take Amazon's example. Using machine learning, the company predicts its customers' needs and views recommendations based on past viewing and order history. In addition, it caters to its key consumers through

Prime Membership that offers greater benefits to members. By prioritising the satisfaction of its top clients, the business enables growth in its brand loyalty.

Many manufacturers are adopting Product as a Service (PaaS) business model to better support their customers' needs. Most commonly, this is seen in the automotive sector where the present population is increasingly focused on usage over ownership. Consider Hyundai's Shopper Assurance programme that is designed to make buying easier. This includes transparent pricing and flexible test drives to ease the purchase process. It also offers a three-day exchange in which if the customer is dissatisfied, the vehicle can be exchanged, or sometimes money is returned to the customer. Consumer preference for sharing economy has given rise to a number of startups in different fields, like Vacatia, LiquidSpace, Turo and Handy, among others.

Smart becoming smarter

Focusing on customer lifecycle is just as important as the entire product journey, if not more. Why connecting to the target audience is important? Because what worked in the past might be obsolete in the future. So, if your audience thinks your product is cool and serves the intended purpose(s), it is cool.

A proper customer relationship management system should be used

by businesses to get an overall view of the end-user experience. An increasing number of companies are utilising AI-enhanced social media platforms, text messaging systems and chatbots for smooth communication with customers. Such startups as Avaamo, Kasisto, Cinnamon and Wysdom are developing AI-powered chatbots.

These technologies allow manufacturers to inform consumers about the benefits of their product(s) and get client feedback, which is necessary to make better products.

The concept of digital twins is also becoming quite popular in manufacturing due to its huge potential. Digital twins, a digital replica of a real-world entity, can be used for improving the process, predictive maintenance and gaining customer insight. Manufacturers can know which features are being liked by most users and where they are getting stuck. In a study of IoT implementation conducted by Gartner, thirteen per cent of organisations implementing IoT projects already used digital twins as of 2018, while 62 per cent were either in the process of establishing digital twins use or planned to do so.

User reviews also help other potential buyers make a decision about whether or not to buy a product. Unifying all these platforms enables quick identification of issues and streamlining of responses. This is useful in providing the required support and adding new functionality to old products through software updates. For instance, Tesla used over-the-air (OTA) software updates in the past to add Auto High Beam feature and Autopilot 2.0 functionality in its cars. Eventually, consumer demands are not only met, these can even be anticipated.

A win-win situation: transformed, meaningful, sustainable products at low cost

A customer-centric decision also means the right financial decision to gain a higher return on investment (ROI). By taking customer insights, manufacturers can save millions on their resulting products as they are able to meet the point-of-demand directly. Even B2B businesses need to understand the needs of the ultimate end-user. According to a press release